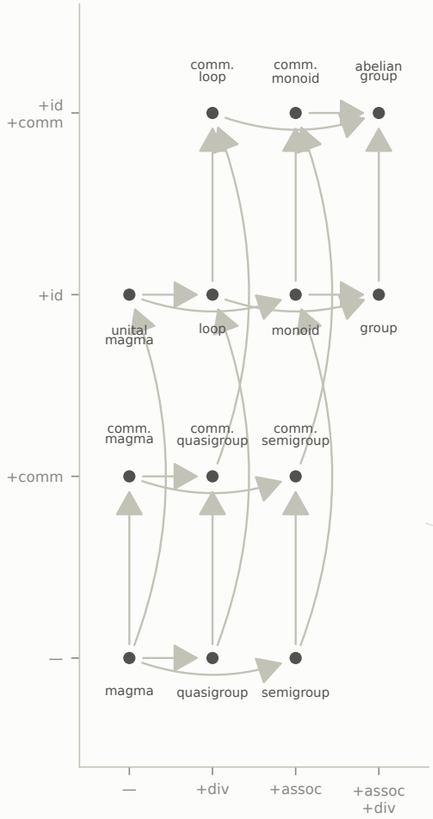
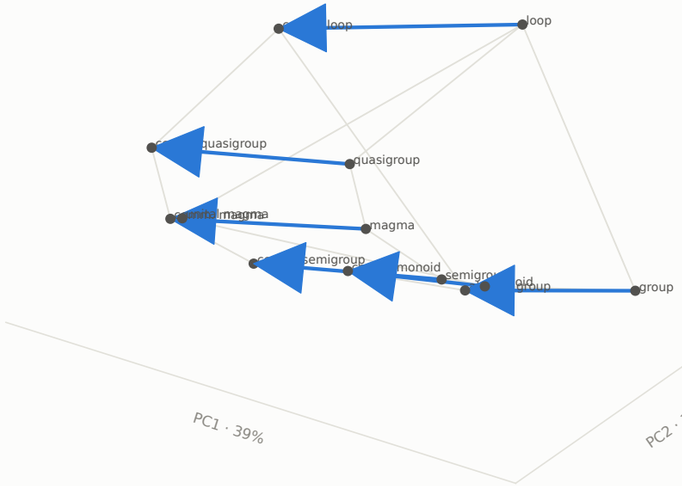
1 · Structures are sets of axioms



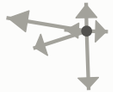
2 · Adding an axiom is an arrow + commutativity in Gemma-4-E2B (top-3 PCs)



3 · Line the arrows up model

consistency 0.74; PC1–3 hold 70% of the variance

random-init twin

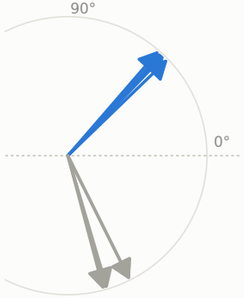


consistency 0.02; PC1–3 hold 55% of the variance

4 · Measure in the full space: angle of each arrow to the others' mean

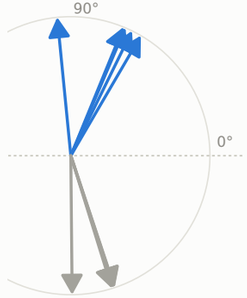
model above the dashed line (blue), twin below (grey); 0° = same direction, 90° = unrelated

+ associativity



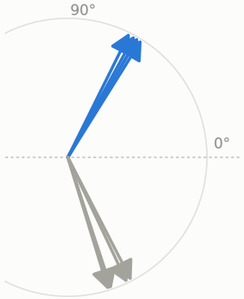
consistency 0.56 (p = 0.002)
twin 0.22

+ identity



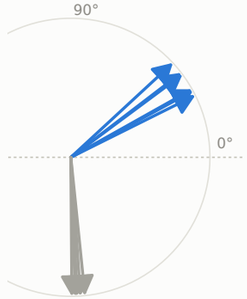
consistency 0.19 (p = 0.090)
twin 0.14

+ divisibility



consistency 0.35 (p = 0.023)
twin 0.23

+ commutativity



consistency 0.74 (p < 0.001)
twin 0.02